

Asadullah Bin Rahman

✉ galib.hstu.cse17@gmail.com · ☎ +8801701905796 · 💻 asadullahgalib007.github.io
🌐 in/asadullah007 · 🔄 AsadullahGalib007 · 🎓 Google Scholar · 🆔 0009-0009-5027-752X

Research Interests: Image Processing, Machine Learning, Quantum Computing

Employment

- 07/25 – Present ♦ **Lecturer, Computer Science and Engineering**
World University of Bangladesh (WUB), Dhaka, Bangladesh
Roles: Undergraduate teaching, thesis & project supervision, competitive programming club mentoring, student counseling
- 07/23 – 07/25 ♦ **Research Assistant**
IoThink Lab, Department of Computer Science and Engineering, HSTU
Focus: Wavelet-based denoising and deep learning for medical image analysis; interdisciplinary dataset collection
Supervisor: Dr. Md. Abdulla Al Mamun 🔗

Education

- 07/23 – 07/25 ♦ **MSc in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Image Denoising using Wavelet Transform and Deep Learning Architectures
CGPA: 3.547/4.00 🔗
- 01/17 – 12/22 ♦ **BSc (Engineering) in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Brain Tumor Classification from Contrast-Enhanced MRI using Deep Learning
CGPA: 3.31/4.00 🔗

Skills

- Quantum Computing ♦ Proficient in Qiskit, Cirq, PennyLane, Classiq; applied experience in Quantum Algorithms, Quantum Machine Learning, Quantum Key Distribution
- Machine Learning ♦ Proficient in NumPy, Pandas, SciPy, Matplotlib, OpenCV, PyTorch, TensorFlow; applied experience in Image Processing, Computer Vision
- Programming ♦ C++, Java, Python; Data Structures, Algorithms, Databases; Solved 300+ problems (CodeForces, HackerRank, UVa)
- Languages ♦ Bangla (Native), English (IELTS Academic: **6.5**, CEFR Level: B2) 🔗
- Others ♦ Linux, Git, LaTeX

Research Publications

Conference Proceedings

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, “Mitigating Noise from Biomedical Images Using Wavelet Transform Techniques,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. 🔗 DOI: [10.1109/ECCE64574.2025.11013062](https://doi.org/10.1109/ECCE64574.2025.11013062).
2. **A. B. Rahman**, M. Touhid Islam, M. R. Islam, M. Sohrawordi, and M. N. Sultan, “Enhanced Brain Tumor Classification from MRI Images Using Deep Learning Model,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. 🔗 DOI: [10.1109/ICCIT60459.2023.10441064](https://doi.org/10.1109/ICCIT60459.2023.10441064).

Preprints

Hyperlinks are embedded in this CV. Please click anywhere with 🔗 for details!

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Deep Learning Architectures for Medical Image Denoising: A Comparative Study of CNN-DAE, CADTra, and DCMIEDNet*, 2025. arXiv: [2508.17223](https://arxiv.org/abs/2508.17223) [eess.IV].  URL: <https://arxiv.org/abs/2508.17223>.
2. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Systematic Evaluation of Wavelet-Based Denoising for MRI Brain Images: Optimal Configurations and Performance Benchmarks*, 2025. arXiv: [2508.15011](https://arxiv.org/abs/2508.15011) [eess.IV].  URL: <https://arxiv.org/abs/2508.15011>.

Teaching

- | | |
|-------------|---|
| Summer 2026 | <ul style="list-style-type: none"> ◇ Course Teacher, WUB. Compiler Design. ◇ Course Teacher, WUB. Object Oriented Programming. ◇ Course Teacher, WUB. Introduction to Programming. |
| Spring 2026 | <ul style="list-style-type: none"> ◇ Course Teacher, WUB. Object Oriented Programming. ◇ Course Teacher, WUB. Introduction to Programming. |
| Fall 2025 | <ul style="list-style-type: none"> ◇ Course Teacher, WUB. Computer Fundamentals. ◇ Course Teacher, WUB. Object Oriented Programming. ◇ Course Teacher, WUB. Computer and Cyber Security. |

Thesis Supervision

- | | |
|-------------|--|
| Summer 2026 | <ul style="list-style-type: none"> ◇ Explainable Deep Learning for Plant Disease Classification Using Convolutional Neural Networks @ WUB: Comparison of CNN architectures for leaf disease classification on PlantVillage crops, with Grad-CAM saliency exposing the leaf regions behind each prediction. ◇ ClassSync Pro: IoT-Based Smart Classroom Automation System @ WUB: LMS-scheduled control of classroom air-conditioning, fans, lighting and projectors with sensor-based occupancy validation; projected 30–40% cut in campus energy use. |
|-------------|--|

Research and Projects

- | | |
|-------|--|
| 02/26 | ◇ Multilayered Security  : Implemented and simulated a research project on Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security. |
| 11/24 | ◇ Guava Fruit Disease Dataset @ IoThink Lab  : Collaborated on dataset collection for interdisciplinary research. |
| 08/24 | ◇ Quantum Machine Learning Projects @ Womanium Program: Four QML projects completed for <i>QML for Conspicuity Detection in Production</i> . <ul style="list-style-type: none"> ○ QML for Anomaly Detection : hybrid quantum-classical model for anomaly detection. ○ Quantvolutional Neural Networks : hybrid quantum convolutional model for MNIST digit classification. ○ Quantum Variational Classifier : quantum classifier for penguin species classification. ○ Quantum Regression Model : learned and predicted the sine function on the interval $[0, 2\pi]$. |

Professional Development

- | | |
|---------------|--|
| 09/24 – 12/24 | ◇ QClass24/25 Fall Semester  : Achieved 96% in 3 ECTS graduate-level program on Quantum Algorithms and QKD. |
| 06/24 – 08/24 | ◇ Womanium Quantum + AI 2024  : Coursework on Quantum Computing and AI. |
| 08/24 | ◇ QGSS 2024 – Quantum Excellence   : Intensive quantum computing boot camp by IBM Qiskit. |

Professional Development (continued)

- ◇ **Classiq Diploma**  : Advanced Quantum Algorithm design.
- ◇ **PennyLane Diploma**  : Quantum Machine Learning Challenge completion.
- 07/24 ◇ **QNickel Diploma**  : Quantum Algorithms workshop.
- ◇ **QBronze Diploma**  : Introductory Quantum Computing and Programming.
- 09/20 ◇ **Python for Everybody Specialization**  : Coursera specialization on Python fundamentals and data structures.

Awards and Achievements

- 06/25 ◇ **Unitary Hack 2025**   : For open source contributions to the quantum software ecosystem.
- 04/25 ◇ **YQuantum 2025**   : Solved BlueQubit's Peaked Circuits Challenge.
- 02/25 ◇ **MIT iQuHACK 2025**  : Solved IonQ's Max Cut problem.
- 08/24 ◇ **Womanium Quantum + AI 2024**  : Program finalist and QSL fellowship nominee.
- 06/24 ◇ **IBM Quantum Challenge 2024**   : Ranked 11th globally.

Volunteering Work

- 04/25 – Present ◇ **Coordinator & Mentor, QBangladesh (QWorld)**  : Mentoring students and promoting quantum computing literacy in Bangladesh.
Hosted Events: QBronze  , QSilver  .

Hobbies

- ◇ Chess 
- ◇ Gardening